Interleaving

Practising how to choose an approach

Use interleaving when students can perform several approaches separately but need practice deciding which one fits. Mix types that require a meaningful choice after initial teaching.

How to do it well

1 Check initial competence. Confirm that students can use each approach in a familiar form with reasonable accuracy.

2 Select contrasting types. Choose tasks for which students must discriminate between related methods or concepts.

3 Mix the sequence. Avoid blocks that reveal which method comes next, while keeping demands manageable.

4 Require the choice. Ask students to identify the approach and explain the clue before or alongside completing the response.

5 Give comparative feedback. Address why an alternative did not fit, then revisit the choice in later mixed practice.

What this sounds like

“These questions need different methods. Before calculating, name the method you would choose and the feature in the question that tells you.”

Classroom example. A Maths practice set mixes tasks requiring different familiar methods. Pupils identify the method and the clue before solving. Feedback compares a plausible alternative with the chosen approach. The difficulty lies partly in selection, whereas a worksheet labelled with one method would have supplied that decision before the pupil began.

Top tips that make the technique work

Teach each approach first. Interleaving should not disguise missing instruction as desirable difficulty.

Remove the topic label. A heading such as “Use this formula” can supply the decision you want students to practise.

Ask for classification before execution. Students can state the approach and the clue that makes it suitable before completing the task.

Mix related alternatives. Contrast methods students might plausibly confuse, rather than assembling a random set.

Expect performance to feel harder. Inspect the nature of errors and provide feedback; difficulty alone is not evidence the sequence is effective.

Do and avoid

Do	Avoid
Practise selection as well as execution.	Mixing untaught content into independent practice.
Keep the approaches individually accessible.	Giving a method cue above every item.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

Barton - Research Spacing and Interleaving